By Team Orange
Old Dominion
University
CS 410
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Patient Path

“From Fax to Fast: Modernizing Medical Record Exchange”

TABLE OF CONTENTS

Who We Are 6-7
Elevator Pitch 8
The Problem 9
Current State 10
Why This Matters 11
Problem Characteristics 12-13
Who is Affected 14
Current Process Flow 15
The Solution 16-20

TABLE OF CONTENTS

Problem Solution Mapping 21
Major Functional Components 22
Patient Path Will. 23
Patient Path Will Not. 24
Risk Assessment Matrix 25
Risk Mitigation Strategies 26
Market Assessment 27
Competition Comparison 28
User Roles 29

TABLE OF CONTENTS

Role Feature Matrix 30
User Stories 31-33
Feature Table 34-36
User Interface Mockups 37-41
Work Breakdown Structure 42
Algorithms 43-49
Database Schemas 50-51
External Interfaces 52
Development Tools 53-59

TABLE OF CONTENTS

Sprint Breakdown 60-61
Conclusion 62
Glossary 63
References 64

Who we are

Kaitlyn Molaison
Project Manager

Kaitlyn is the project manager of PatientPath, contributing to design, development, and documentation while helping guide the team's overall vision and progress.

James Woodson
Software Developer

James is a PatienPath team member who contributes to software design implementation as well as other backend tasks.

Darnell Barnes
Full Stack Developer

Darnell is a PatientPath team member who contributes to the frontend user interface and backend system functionality.

Who we are

Grace Wright
Front End Developer

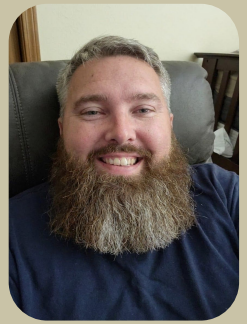
Grace is a PatientPath team member responsible for the front end development. She will contribute to the front end design of the website, and ensure that it is working properly for user's.

Christopher DeHaven
Software Developer

Christopher is a developer for PatientPath who contributes to software design implementation and improving functionality and efficiency within the system.

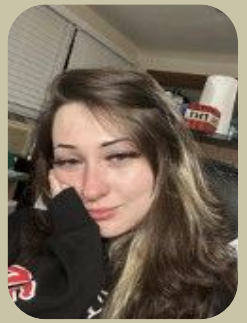
Stephen Garrison
Software Developer

Stephen Garrison is a team member on PatientPath who contributes to system design, technical writing, and building clear project communication through the team website.



Emily Nowak
Documentation Specialist

Emily is a PatientPath team member responsible for documentation and creating process flow diagrams to clearly communicate system design and project goals.



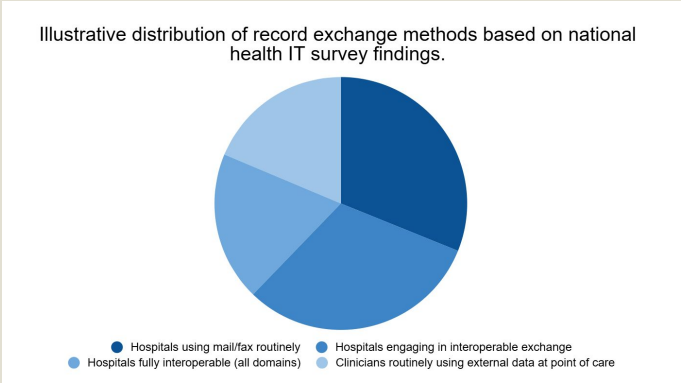
Elevator Pitch

PatientPath is a secure web-based platform that replaces outdated fax systems in healthcare with instant digital record sharing. It allows providers to access and share patient records in real time while integrating with existing EHR (electronic health record) systems. By reducing delays, errors, and administrative work, PatientPath helps deliver faster, safer, and more coordinated care, so patients’ medical stories follow them wherever they go.

The Problem

Current medical transfer methods are inefficient and slow.

- 65% of all transfers rely on fax machines despite the availability of digital alternatives
- Slow, unreliable, and prone to errors
  - Issues include lost pages, unreadable documents, and delayed delivery
- Direct impact on patient care
  - Leads to postponed diagnosis and repeated tests
- Disproportionate Harm
  - Especially impactful in rural and underfunded facilities



Office of the National Coordinator for Health Information Technology. (2023). *Interoperable exchange of patient health information among U.S. hospitals: 2023 data brief.* U.S. Department of Health & Human Services. https://www.healthit.gov/data/data-briefs/interoperable-exchange-patient-health-information-among-us-hospitals-2023

Current State of Healthcare IT

Although some healthcare practices have implemented EHR systems, *interoperability still remains a major challenge*.
Key issues include:

- Many EHR systems do not communicate across different organizations.
- Healthcare IT systems remain siloed.
- Smaller and rural communities often lack modern infrastructure.
- Regulatory requirements (HIPAA) increase complexity.

As a result, when systems cannot exchange data directly, providers often default to manual or outdated communication methods.

Source: Office of the National Coordinator for Health Information Technology (2023). Interoperable exchange of patient health information among U.S. hospitals.

Why This Matters

Healthcare is very time sensitive. Delays in information transfer can lead to:

- Postponed specialist appointments
- Repeated diagnostic testing
- Incomplete patient histories
- Increased workload for administrative staff
- All around frustration for patients and providers

In an industry where minutes can affect outcomes, inefficiencies in information exchange have real clinical and financial consequences.

Problem Characteristics

Patient Privacy Risk

Outdated security protocols

Confidential data vulnerability

Disconnected Systems

Limited interoperability

Inefficient Transfer Process

Record delays

Document loss

Illegible fax transmissions

Problem Characteristics

Scalability Barriers

Limited growth adaptability

Rigid architecture

Poor Usability

Cluttered interfaces

Poor navigation

Lack of user roles

Connectivity Constraints

Online-only architecture

Downtime vulnerability

Who is Affected?

- Patients
- Providers
- Admin Staff
- Rural Facilities
- Families/Caregivers

Current Flow Process

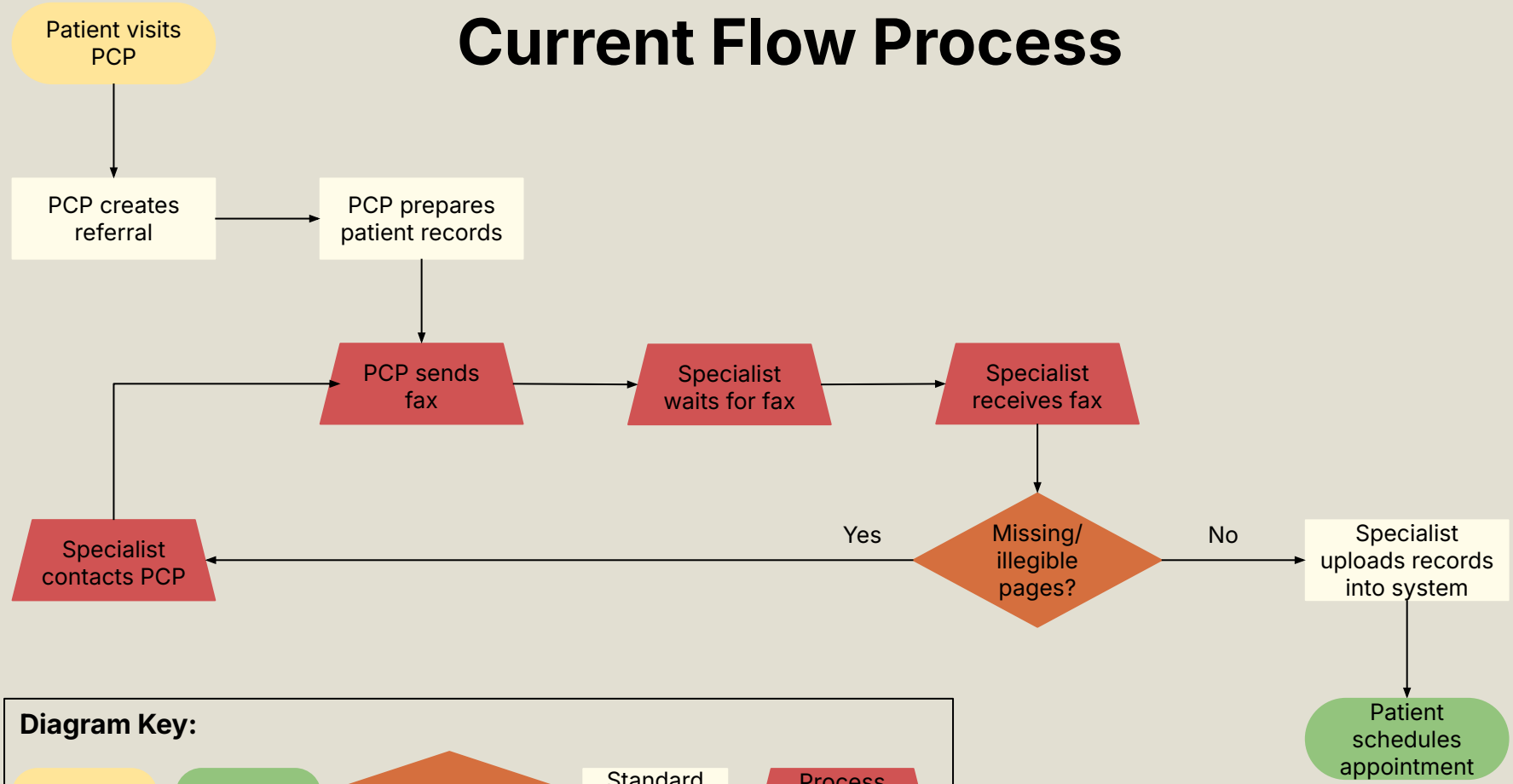


Diagram Key:

Start | Stop | Decision | Standard Step | Process Bottleneck

The Solution

PatientPath is a secure, centralized digital platform designed to replace outdated fax-based communication in healthcare.

Instead of faxing documents, PatientPath enables providers to instantly upload, access, and share medical records through a secure web-based system.

The platform supports hospitals, clinics, private practices, healthcare networks, and their patients.

Unlike traditional EHR systems, PatientPath focuses on secure interoperability and cross-organization communication. It acts as a **secure bridge between systems**, reducing reliance on fax while improving cybersecurity, compliance, and usability.

Solution Characteristics

Security & Compliance

- End-to-end encryption (data in transit + at rest)
- HIPAA-aligned architecture
- Role-Based Access Control (organization-level permissions)
- Multi-factor authentication
- Full audit trail logging

Solution Characteristics

Interoperability & Architecture

- API-first REST architecture
- FHIR & HL7 compatibility
- EHR-agnostic integration (Epic, Oracle, etc.)
- Plugin-based deployment model

Solution Characteristics

Performance & Scalability

- Cloud-hosted (AWS/Azure/GCP compatible)
- 99.9%+ uptime target
- Horizontal scaling architecture
- Offline sync resilience

Solution Process Flow

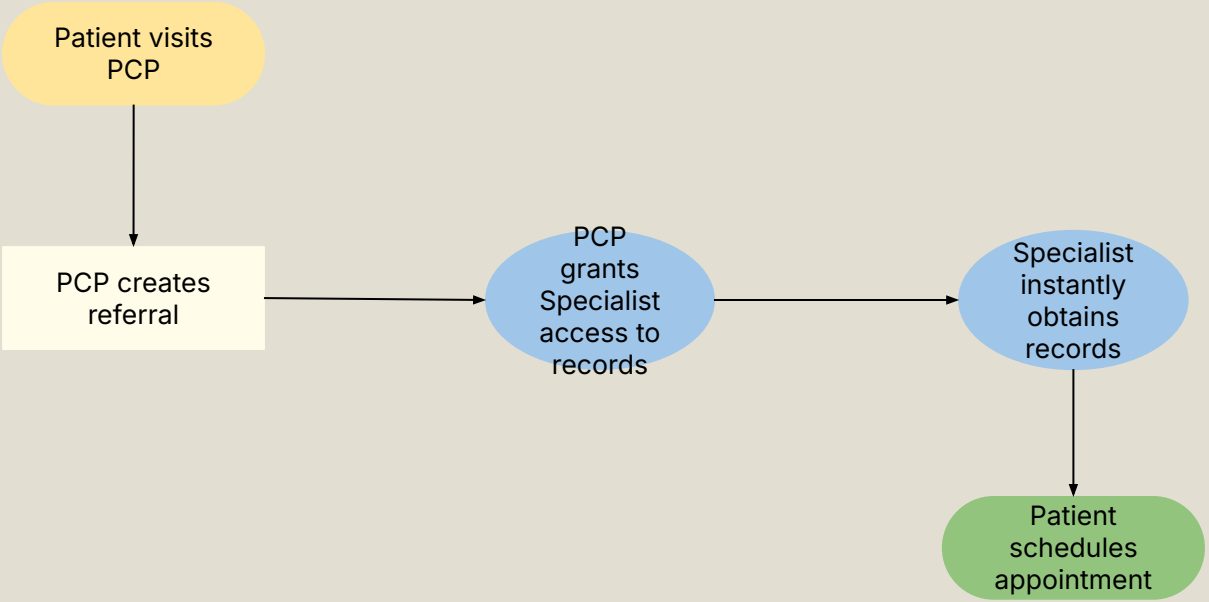


Diagram Key:

Problem–Solution Mapping

	Identified Problem	System Feature	Expected Impact
1	Reliance on fax-based record transfers	Secure real-time digital document exchange	Reduced delays & elimination of document loss
2	Limited interoperability between EHR systems	API-first architecture with FHIR/HL7 compatibility	Standardized cross-organization data exchange
3	Manual referral tracking & follow-up processes	Automated referral workflow with status monitoring	Improved efficiency & faster referral completion
4	Communication methods withsecurity vulnerabilities	End-to-end encryption with RBAC and MFA	Enhanced HIPAA-aligned compliance & access control
5	Connectivity instability in rural / low-resource areas	Offline caching with automatic synchronization	Sustained usability in low-bandwidth environments

Major Functional Components Architecture

Presentation Layer

- Provider Dashboard
- Patient Portal
- Admin Console
- Login & MFA Screens
- Notification Interface

USERS

Patients

Providers

Administrators

Application Layer

- *API Gateway / Web Server*
- Identity & Access Management
- Patient Record Management
- Secure Record Exchange Service
- FHIR / HL7 Integration Service

Data Layer

- PostgreSQL Database (user & organization)
- Encrypted Document Storage (Medical Records)
- Audit Logs
- Consent Records
- FHIR Normalized Data

Patient Path Will

What Patient Path Will Do

- Enable secure digital transfer of medical records between providers
- Provide a centralized hub for accessing and managing patient documents
- Offer real-time tracking of record requests and transfer status
- Support digital patient consent and authorization workflows
- Reduce errors with automated validation and completeness checks
- Maintain HIPAA-compliant audit trails and access controls

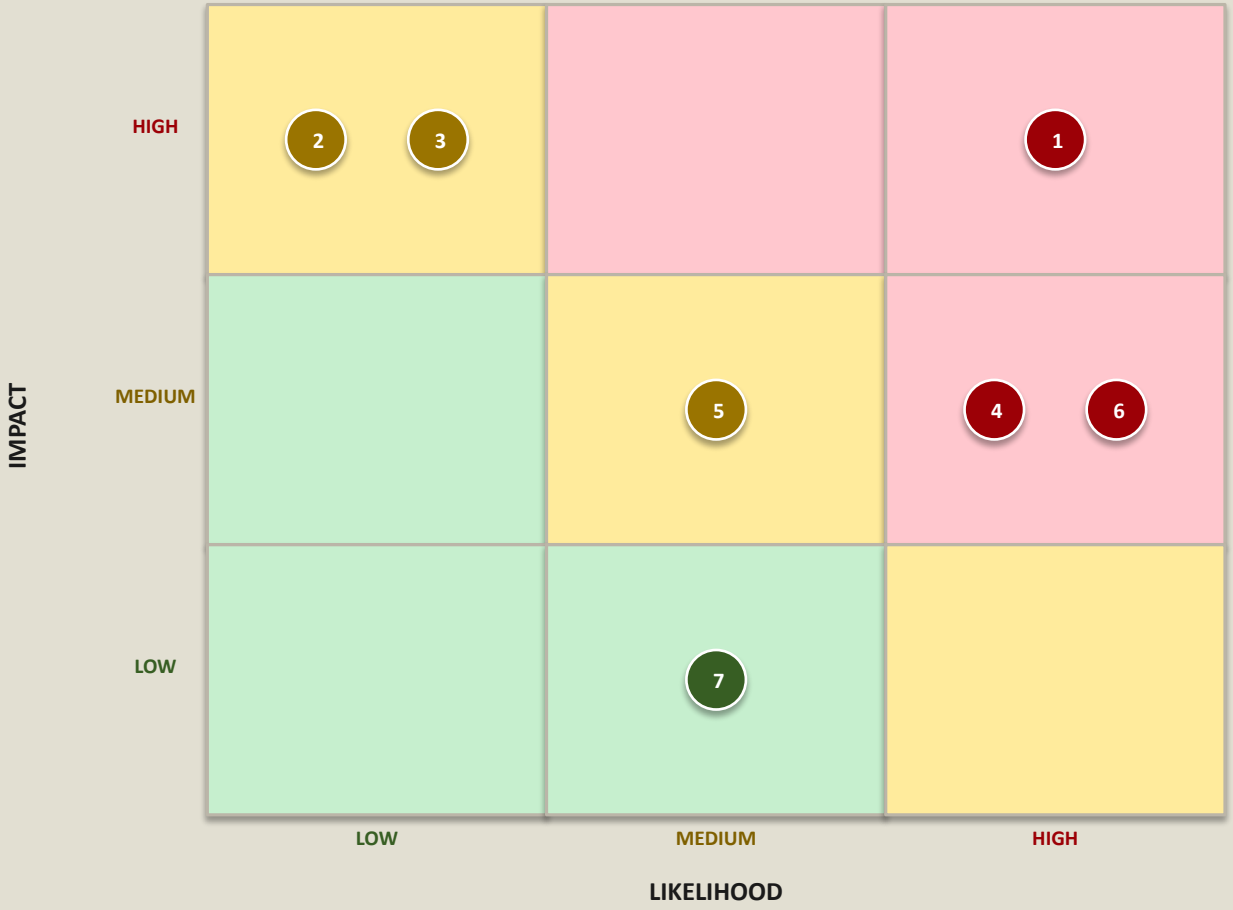
Patient Path Will Not

Provide medical diagnoses or clinical recommendations - The platform does not assist in medical decision making. It’s role is limited to securely transferring and accessing information.

Handle hospital management functions - Scheduling, billing, and internal hospital operations are already handled by specialized systems.

Function as an emergency communication system - The platform supports data exchange, not urgent/emergency situations.

Risk Assessment Matrix



Risk Legend

HIGH	MED	LOW
Technical	Customer	Legal/Sec

1. EHR Integration Complexity
   Technical • Control
2. Cybersecurity / Data Breach
   Legal & Sec • Control
3. HIPAA Non-Compliance
   Legal & Sec • Avoid
4. Provider Adoption Resistance
   Customer • Control
5. System Downtime / Cloud Failure
   Technical • Transfer
6. Scope Creep
   Technical • Avoid
7. Patient Trust / Privacy Concerns
   Customer • Monitor

Risk Mitigation Strategies

Organized by risk category | Approach types: Control · Avoid · Transfer · Monitor

Technical Risks

1 EHR Integration Complexity

Control

Phased FHIR/HL7 rollout with sandbox testing before production deployment

5 System Downtime / Cloud Failure

Transfer

SLA with cloud provider (AWS/Azure) requiring 99.9%+ uptime; failover redundancy

6 Scope Creep

Avoid

Strict stage-gate process with weekly milestone reviews and frozen requirements

Customer Risks

4 Provider Adoption Resistance

Control

Structured onboarding program, intuitive UI design, and workflow alignment workshops

7 Patient Trust / Privacy Concerns

Monitor

Transparent consent flows, clear privacy policy, and regular patient communication

Legal & Security Risks

2 Cybersecurity / Data Breach

Control

End-to-end encryption, MFA, annual penetration testing & incident response plan

3 HIPAA Non-Compliance

Avoid

Legal review at each dev phase; automated compliance checks built into CI/CD pipeline

Market Assessment

Target Segments

- Hospitals and Healthcare networks
- Independent private practices
- Specialty Clinics

Competitive Landscape

- Traditional fax transfers
- Electronic Health Record systems (Epic, Oracle Healthcare, and eClinicWorks)

Competition Comparison Matrix

Capability	PatientPath	Traditional Fax	Kno2 (Digital Fax)	Redox (Healthcare APIs)	CommonWell HIE
Real-time record transfer	✓	✗	Limited	✓	✓
Cross-organization data exchange	✓	✓ (manual)	✓	✓	✓
EHR interoperability (FHIR/HL7)	✓	✗	Limited	✓	✓
Automated referral workflow	✓	✗	Limited	✗	✗
Secure audit logging	✓	✗	✓	✓	✓
Offline support for rural clinics	✓	✓	✗	✗	✗
Ease of deployment	✓	✓	✓	Limited	Limited

User Roles

Primary Users of the PatientPath System

Healthcare Providers

PCP (initiates referrals) | Specialist (receives records)

- Upload and share patient medical records
- Access records from other organizations in real-time
- Initiate and track referrals
- Grant and manage access permissions

Administrative Staff

Record & Workflow Management

- Manage patient documentation and record requests
- Monitor referral workflows and status updates
- Ensure completeness of transferred records
- Handle communication between organizations

Patients

Read-Only Access & Consent

- Provide consent for record sharing
- View transfer status of their medical records
- Access personal health documents (read-only)
- Maintain control over who can access their data

System Administrators

Security, Compliance & Configuration

- Manage user accounts and roles (RBAC)
- Monitor system security and audit logs
- Configure integrations with EHR systems
- Ensure compliance with HIPAA and security policies

Role–Feature Matrix

Which roles interact with which system features

Feature	PCP	Specialist	Admin Staff	Patient	Sys Admin
Secure Record Upload	✓	✓	✓	✗	✗
Real-Time Sharing	✓	✓	monitor	status	✗
Referral Tracking	✓	view only	✓	status	✓
Patient Consent Mgmt	request	✗	log only	✓	✓
RBAC / Audit Logging	subject	subject	✓	✗	✓
Offline Sync	✓	✓	✓	✗	✓
MFA Login	✓	✓	✓	✓	✓

✓ **Full Access** ✗ **No Access** *Italics = limited / read-only / role-specific access*

User Stories — Upload & Referral Workflow

Workflows: Record Upload · Referral Initiation · Status Tracking · Validation

Healthcare Provider · US-1 Secure Record Upload

As a healthcare provider, I want to upload patient records securely so that I can share them with other providers in real time.

Acceptance Criteria:

- File is encrypted on upload; only authorized users may access records
- Upload completes within the defined time threshold
- System validates file format and completeness before submission
- Provider receives on-screen confirmation after successful upload

→ Trace: Provider Dashboard (upload form) → ValidateAndEncrypt module → Encrypted Doc Storage + PostgreSQL → Dashboard (confirmation)

Healthcare Provider · US-2 Referral Initiation

As a healthcare provider, I want to initiate a referral and grant record access in a single workflow so that I can eliminate manual faxing.

Acceptance Criteria:

- Referral created with one form submission
- Access granted to receiving organization immediately upon submission
- Provider receives confirmation notification
- Incomplete packages are blocked with specific error messages

→ Trace: Provider Dashboard (referral form) → CreateReferral module → PostgreSQL + IAM → Specialist Dashboard (notification)

Administrative Staff · US-3 Referral Status Monitoring

As an administrative staff member, I want to monitor the status of all active referrals so that I can eliminate manual phone follow-ups.

Acceptance Criteria:

- Dashboard displays real-time status for all active referrals
- Status updates within 30 seconds of any change
- Staff can filter referrals by provider, date, or status
- Completed referrals are archived and remain searchable

→ Trace: Admin Console (dashboard) → TrackReferralStatus module → PostgreSQL → Admin Console (real-time status update)

Administrative Staff · US-4 Incomplete Record Flag

As an administrative staff member, I want the system to flag incomplete record packages before transmission so that referrals are not rejected.

Acceptance Criteria:

- System validates required fields before allowing transmission
- Incomplete packages are blocked with a specific error message
- Staff can resolve issues and resubmit without starting over
- Validation runs automatically on every save action

→ Trace: Provider Dashboard (pre-send check) → ValidateRecord module → API Gateway → Dashboard (error message + resubmit prompt)

User Stories — Consent & Monitoring Workflow

Workflows: Patient Consent · Transfer Visibility · Stale Referral · Offline Sync

Patient · US-9 Digital Consent

As a patient, I want to provide digital consent for record sharing so that I maintain control over who accesses my health data.

Acceptance Criteria:

- Consent workflow is accessible from the patient portal
- Consent is timestamped, logged, and auditable
- Patient can revoke access at any time
- Provider is notified immediately upon revocation

→ Trace: Patient Portal (consent form) → ManageConsent module → Consent Records store → Portal (confirmation) + Provider Dashboard (access update)

Patient · US-10 Transfer Status View

As a patient, I want to view the real-time transfer status of my records so that I am informed about my referral progress.

Acceptance Criteria:

- Portal shows current status in plain, non-technical language
- Status reflects changes within 60 seconds
- Historical transfers are visible and date-stamped
- Patient receives a notification when records are accessed

→ Trace: Patient Portal (status screen) → TrackReferralStatus module → PostgreSQL → Patient Portal (real-time status update)

Administrative Staff · US-8 Stale Referral Alert

As an administrative staff member, I want the system to alert me when a referral is pending without action for 48 hours so that no patient falls through the cracks.

Acceptance Criteria:

- System monitors all active referrals for inactivity
- Alert sent to assigned admin after 48 hours of no update
- Admin can mark as reviewed to reset the inactivity timer
- Unresolved items are escalated to supervisor after 72 hours

→ Trace: Admin Console → TrackReferralStatus module → PostgreSQL → Admin Console (stale referral alert + escalation trigger)

Healthcare Provider · US-5 Offline Record Access

As a healthcare provider, I want to access cached records during an outage so that rural clinic downtime does not interrupt patient care.

Acceptance Criteria:

- Records accessed during a session are cached locally
- Cached data remains accessible during a full network outage
- Queued writes sync automatically within 60 sec of reconnection
- No data loss occurs during sync; server state resolves conflicts

→ Trace: Provider Dashboard (offline indicator) → CacheRecords + SyncQueue modules → Local cache → PostgreSQL → Dashboard (sync confirmation)

User Stories — Security & Validation Workflow

Workflows: Transfer Rejection · Access Control · Audit Trail · RBAC

Healthcare Provider · US-6 Invalid Transfer Rejection

As a healthcare provider, I want the system to reject invalid or incomplete transfers so that corrupt or partial data never reaches the receiving provider.

Acceptance Criteria:

- System blocks transmission of files failing format validation
- Error message specifies which fields or files are missing
- No partial data is transmitted under any condition
- Provider must resolve all errors before resubmission

→ Trace: Provider Dashboard (submit) → ValidateAndEncrypt module → Encrypted Document Storage → Provider Dashboard (validation error + field detail)

System Administrator · US-7 Unauthorized Access Alert

As a system administrator, I want unauthorized access attempts to trigger an automatic alert so that potential breaches are detected promptly.

Acceptance Criteria:

- Three failed login attempts lock the account automatically
- Admin receives an alert within 30 seconds of account lockout
- All attempts are logged with IP address and timestamp
- Account unlock requires MFA verification by the user

→ Trace: Login & MFA Screens → Identity & Access Management → LogAccessEvent module → Audit Logs + PostgreSQL → Admin Console (alert)

System Administrator · US-11 Immutable Audit Trail

As a system administrator, I want all data access events logged in an immutable audit trail so that HIPAA compliance requirements are met.

Acceptance Criteria:

- Every access, transfer, and login event is logged
- Logs include timestamp, user ID, and action type
- Logs cannot be edited or deleted by any user
- Audit reports are exportable in CSV and PDF format

→ Trace: All system actions → LogAccessEvent module → Audit Logs DB (append-only) → Admin Console (compliance report export)

System Administrator · US-12 RBAC Management

As a system administrator, I want to manage user roles through RBAC so that access is restricted to authorized personnel only.

Acceptance Criteria:

- Roles can be assigned, modified, and revoked in real time
- Permission changes take effect immediately upon save
- Unauthorized access attempts trigger an automated alert
- Role history is logged for each user account

→ Trace: Admin Console (role management UI) → Identity & Access Management → PostgreSQL (user roles) → Admin Console (confirmation + audit log entry)

Feature Table (1 of 3) — User-Facing: Core Functionality

User-Facing Features

★ = Novel / Differentiating Feature US = User Story number

Feature	Description	UI Screen	DB / API	Roles	Maps To	★
Core User-Facing Functionality						
Secure Record Upload	Files are validated for format and completeness before upload. Invalid files are blocked at the API Gateway. Approved records are encrypted and routed to Encrypted Document Storage. Upload completes within 5 seconds.	*Provider Dashboard (upload form)*	*Encrypted Doc Storage PostgreSQL (records)*	Provider Admin	**US-1, US-6**	
Real-Time Record Sharing	IAM issues a scoped, time-limited access token to the receiving org upon referral creation. Provider and Specialist dashboards update instantly via notification. No manual steps required from either party.	*Provider Dashboard Specialist Dashboard*	*IAM (token service) PostgreSQL (referrals)*	Provider	**US-1, US-2**	★
Referral Tracking System	Single-form referral initiation with pre-send completeness validation. Status transitions (Pending → In Review → Accepted → Complete) stored in PostgreSQL and pushed to dashboards in real time. Referrals inactive for 48hr trigger admin alerts.	*Provider Dashboard Admin Console*	*PostgreSQL (referrals) API Gateway*	Provider Admin	**US-2, US-3 US-4, US-8**	★
Patient Consent Management	Structured consent form in the Patient Portal. Signed consents are timestamped and stored in the Consent Records store. Revocations immediately propagate to all active IAM sessions, revoking access tokens in real time.	*Patient Portal (consent form)*	*Consent Records store IAM (session revocation)*	Patient Provider	**US-9**	★

Feature Table (2 of 3) — User-Facing: Visibility, Auth & Offline

User-Facing Features

★ = Novel / Differentiating Feature US = User Story number

Feature	Description	UI Screen	DB / API	Roles	Maps To	★
Visibility, Authentication & Offline Resilience						
Transfer Status View	Patient Portal displays real-time referral status in plain, non-technical language. Historical transfers are visible and date-stamped. Patient receives a notification when their records are accessed by a provider.	*Patient Portal (status screen)*	*PostgreSQL (referrals)* *API Gateway*	Patient	US-10	
Centralized Record Access	Unified cross-organization view of patient records via the API Gateway. All records are normalized to FHIR R4 format for consistent display regardless of the originating EHR system.	*Provider Dashboard* *Patient Portal*	*Encrypted Doc Storage* *FHIR Normalized Data*	Provider Patient	US-1, US-10	
Multi-Factor Authentication	TOTP two-step verification is required on all accounts at login. Three consecutive failed attempts trigger automatic account lockout and an admin alert within 30 seconds. Unlock requires MFA re-verification by the user.	*Login & MFA Screens*	*PostgreSQL (users)* *IAM*	All	US-7	
Offline Sync Capability	Recently accessed records are cached locally during network outages. Queued writes are held in a local sync queue and reconciled with the server within 60 seconds of reconnection. Server state is the source of truth for conflict resolution.	*Provider Dashboard (offline mode indicator)*	*Local cache (SW)* *PostgreSQL (sync queue)*	Provider	US-5	★

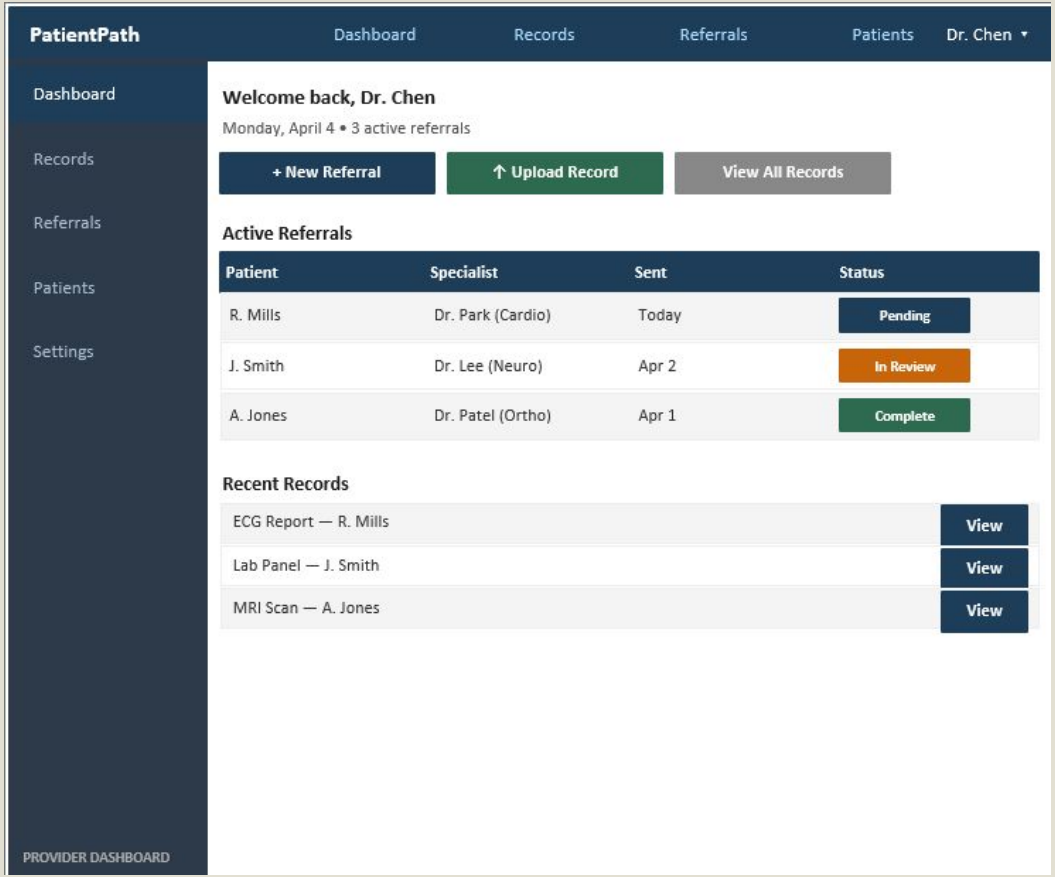
Feature Table (3 of 3) — System & Architecture Components

Infrastructure & Integration

★ = Novel / Differentiating Feature US = User Story number

Feature	Description	UI Screen	DB / API	Roles	Maps To	★
Security & Compliance						
End-to-End Encryption	AES-256 at rest; TLS 1.3 in transit. Keys rotate every 90 days via cloud KMS.	*N/A — infrastructure layer*	*API Gateway (TLS)* *Encrypted Doc Storage*	All	**US-1, US-11**	
Role-Based Access Control	Four roles enforced per API request by IAM. Permission changes propagate instantly; violations trigger alerts.	*Admin Console (role management)*	*IAM* *PostgreSQL (user roles)*	Admin Sys Admin	**US-7, US-12**	
Audit Logging	Every access, transfer, and login logged with timestamp, user ID, IP, and action. Exportable as CSV/PDF.	*Admin Console (audit reports)*	*Audit Logs DB (append-only)*	Admin Sys Admin	**US-11**	
Interoperability						
FHIR / HL7 Integration	Maps EHR data to FHIR R4 via configurable layer. Supports RESTful endpoints and HL7 v2.x queuing.	*N/A — integration layer*	*FHIR Normalized Data* *API Gateway*	Sys Admin	**US-1, US-2**	★
EHR Integration Layer	Plugin adapters for Epic, Oracle, and eClinicalWorks — each handles auth, extraction, and translation independently.	*N/A — integration layer*	*API Gateway* *External EHR APIs*	Sys Admin	**US-1, US-2**	★
API Gateway	Routes all traffic; handles TLS termination, request routing, rate limiting, and token validation.	*N/A — infrastructure layer*	*All internal services (routing layer)*	Sys Admin	**US-1, US-2** **US-11**	
Performance & Reliability						
Cloud-Based Infrastructure	Scalable on AWS, Azure, or GCP with auto-scaling and IaC. Targets 99.9%+ uptime per SLA.	*N/A — infrastructure layer*	*AWS / Azure / GCP* *IaC (Terraform)*	Sys Admin	**US-5, US-11**	
High Availability Architecture	Active-passive failover with DB replication and load balancing. 99.9%+ uptime SLA; no single point of failure.	*N/A — infrastructure layer*	*PostgreSQL (replica)* *Load Balancer*	All	**US-5, US-11**	

User Interface Mockups : Provider Dashboard



Design Annotations

User Stories
- US-1 (Secure Upload)
- US-2 (Referral Init.)
- US-3 (Status Monitor)

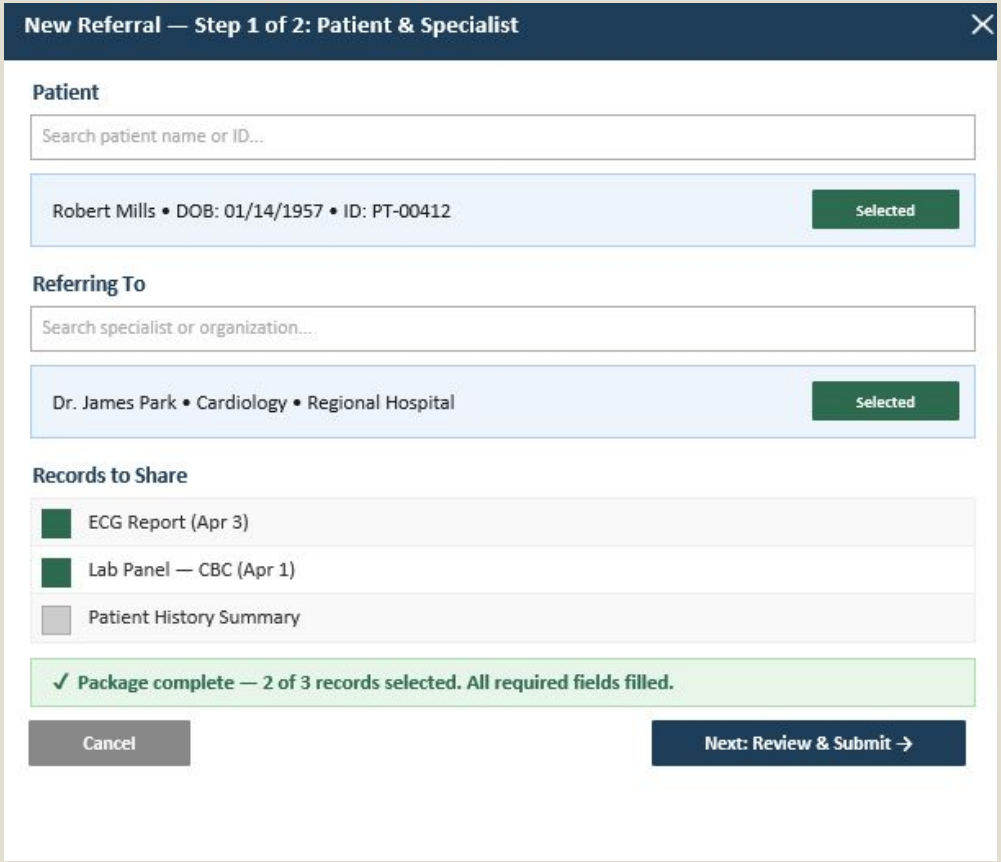
Features
- Real-time Record Sharing
- Referral Tracking System
- Centralized Record Access

Algorithms
- CreateReferral
- TrackReferralStatus
- ValidateAndEncrypt

CRUD Operations
- CREATE referral
- READ records/referrals
- UPDATE referral status

DB Tables/APIs
- Referrals
- Patients
- Records
- Organizations

User Interface Mockups : Referral Management



Design Annotations

User Stories
- US-2 (Referral Init.)
- US-4 (Record Flag)
- US-1 (Secure Upload)

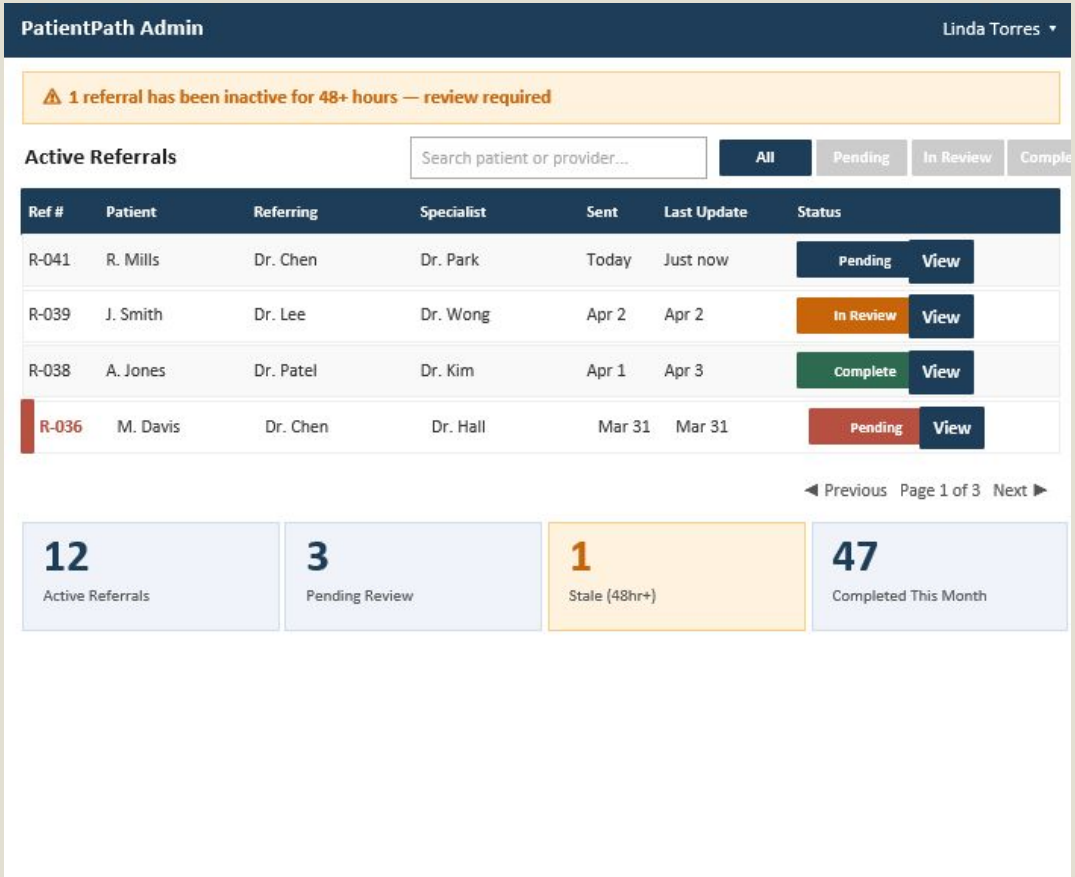
Features
- Real-time Record Sharing
- Referral Tracking System
- Secure Record Upload

Algorithms
- CreateReferral
- ValidateAndEncrypt

CRUD Operations
- CREATE referral
- READ records/referrals
- READ records
- CREATE access_token

DB Tables/APIs
- Referrals
- Patients
- Providers
- Records
- iam_tokens

User Interface Mockups : Admin Console



Design Annotations

User Stories
- US-3 (Status Monitor)
- US-4 (Record Flag)
- US-1 (Record Flag)

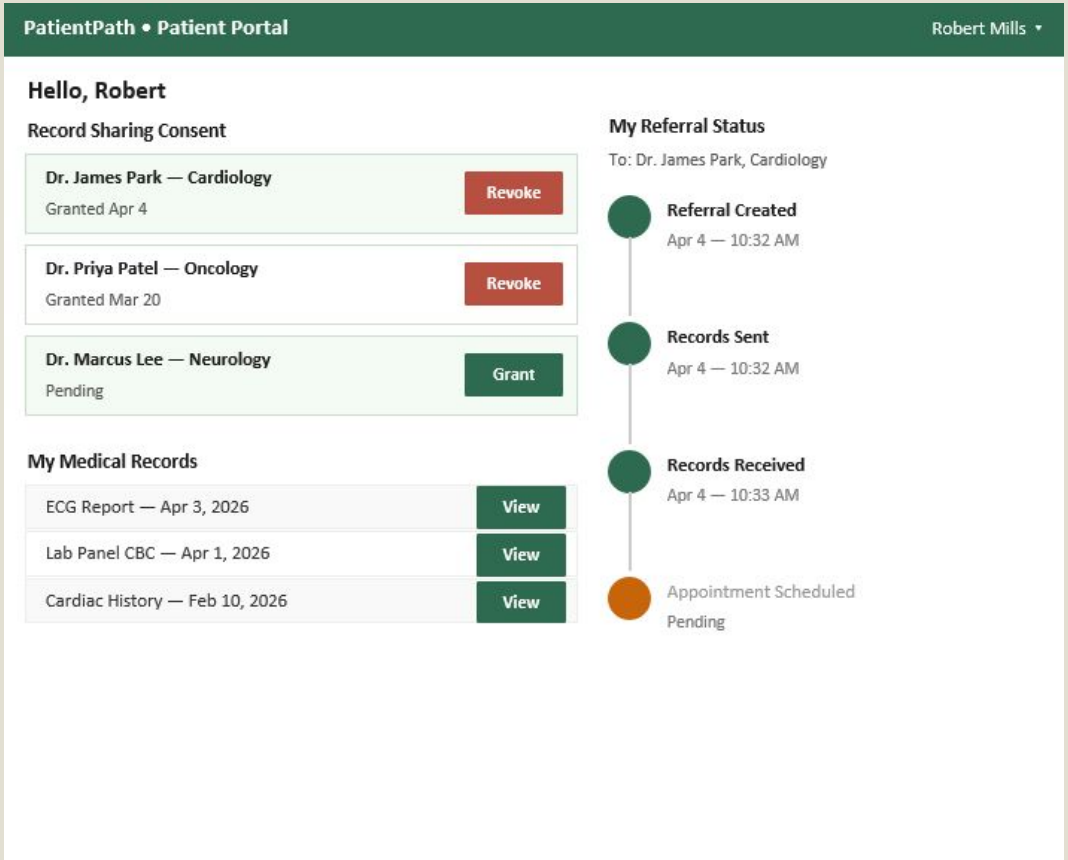
Features
- Audit Logging
- Referral Tracking System

Algorithms
- TrackReferralStatus

CRUD Operations
- CREATE referral (filtered)
- UPDATE referral status
- READ audit_logs

DB Tables/APIs
- Referrals
- Audit_logs
- Users
- Patients

User Interface Mockups : Patient Portal pt1



Design Annotations

User Stories
- US-9 (Digital Consent)
- US-10 (Transfer Status)

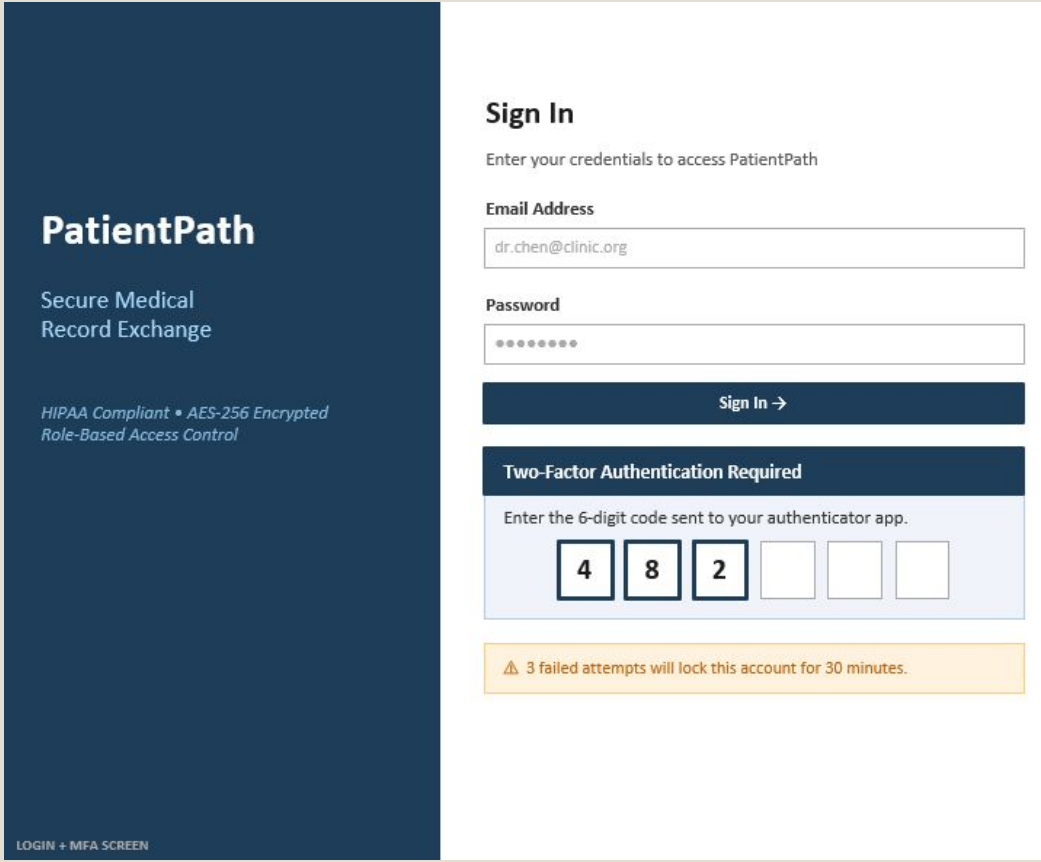
Features
- Patient Consent Managment
- Centralized Record Access

Algorithms
- ManageConsent

CRUD Operations
- CREATE conset record
- DELETE/REVOKE consent
- READ records
- READ referral status

DB Tables/APIs
- Consent_records
- Patients
- Referrals
- records

User Interface Mockups : Patient Portal pt2



Design Annotations

User Stories
- US-7 (Unauthorized Access Alert)

Features
- Multi-Factor Authentication
- Role-Based Access Control
- Audit Logging

Algorithms
- LogAccessEvent

CRUD Operations
- READ user credentials
- CREATE session/token
- CREATE audit_logentry

DB Tables/APIs
- Users
- Sessions
- Audit_logs
- roles

Work Breakdown Structure

PATIENT PATH SYSTEM

1. User Interface Design

1.1 Research & Planning
- Identify user roles
- Review user stories
- Define UI requirements

1.2 Wireframing
- Provider Dashboard
- Referral Form
- Admin Console
- Patient Portal

1.3 Mockup Design
- High-fidelity mockups
- Layout consistency
- Navigation design

1.4 UI Testing & Feedback
- Review with team
- Revise mockups
- Finalize UI screens

2. Algorithm Design

2.1 Referral Processing Logic
- Referral validation
- Status tracking
- Referral routing

2.2 Security & Access Control
- Role-based access
- Authentication
- Authorization checks

2.3 Data Validation
- File validation
- Input validation
- Error handling

2.4 Notification System
- Alert generation
- Status notifications
- Expiration alerts

3. Database Design / Schema

3.1 Identify Entities
- User
- Patient
- Referral
- Medical Record
- Consent / Org

3.2 Define Relationships
- User↔Org
- Patient↔Referral
- Referral↔MedRec
- Referral↔Consent

3.3 Tables & Attributes
- Primary keys
- Foreign keys
- Constraints

3.4 Schema Validation
- CRUD mapping
- Relationship validation

4. Data Management Approach

4.1 Storage Strategy
- Database selection
- Data structure planning

4.2 Data Security
- Encryption
- Secure file storage
- Access control

4.3 Data Access & Retrieval
- API endpoints
- Query optimization
- Data filtering

4.4 Data Maintenance
- Backup strategy
- Data cleanup
- Logging & auditing

Algorithm 1: ValidateAndEncrypt

ValidateAndEncrypt

User Stories: US-1, US-6

Inputs

- File upload request
- User auth token
- Record metadata

Outputs

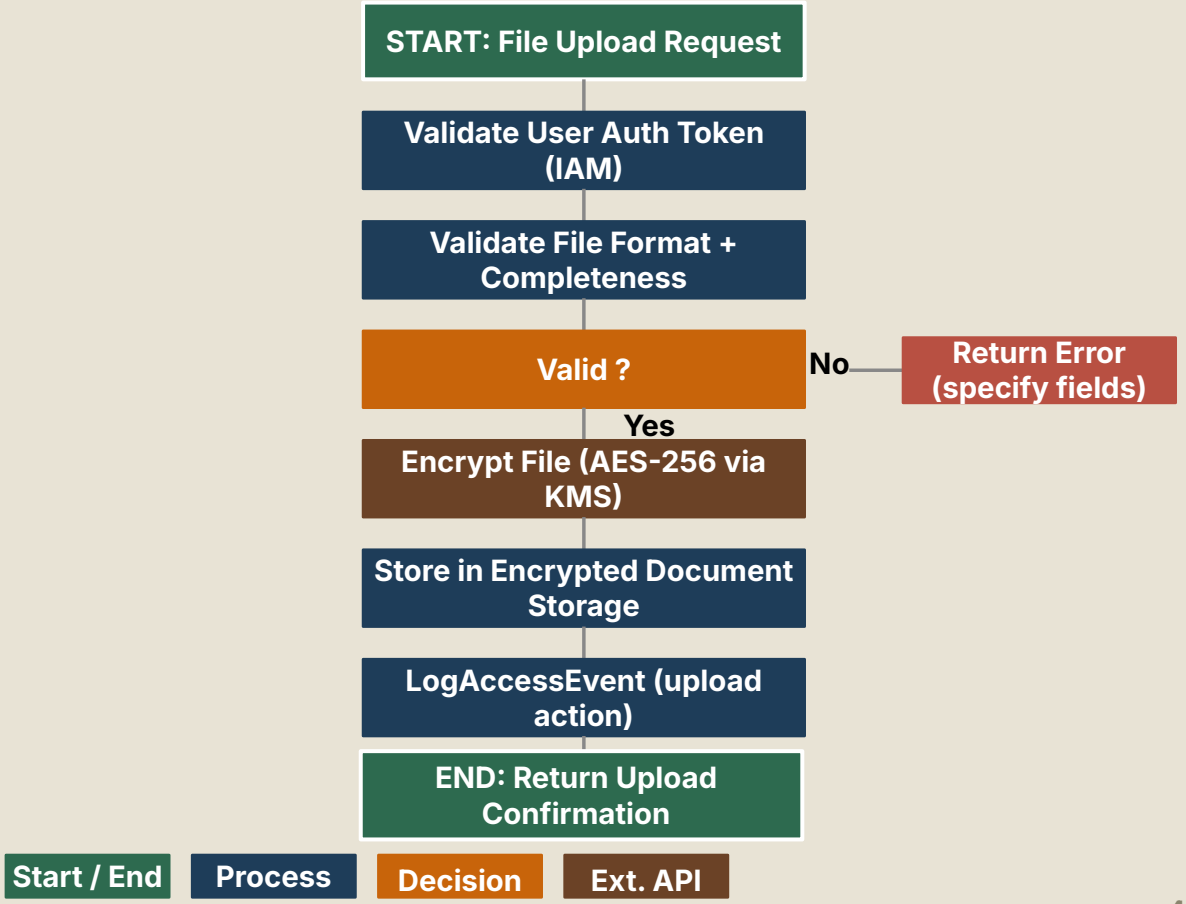
- Encrypted file in Document Storage
- Upload confirmation
- Audit log entry

External APIs

- AWS KMS / Azure Key Vault (AES-256 key management)

DB Tables

- Records (encrypted upload), Audit_Logs (every access event)



Algorithm 2: CreateReferral

CreateReferral

User Stories: US-2, US-3

Inputs

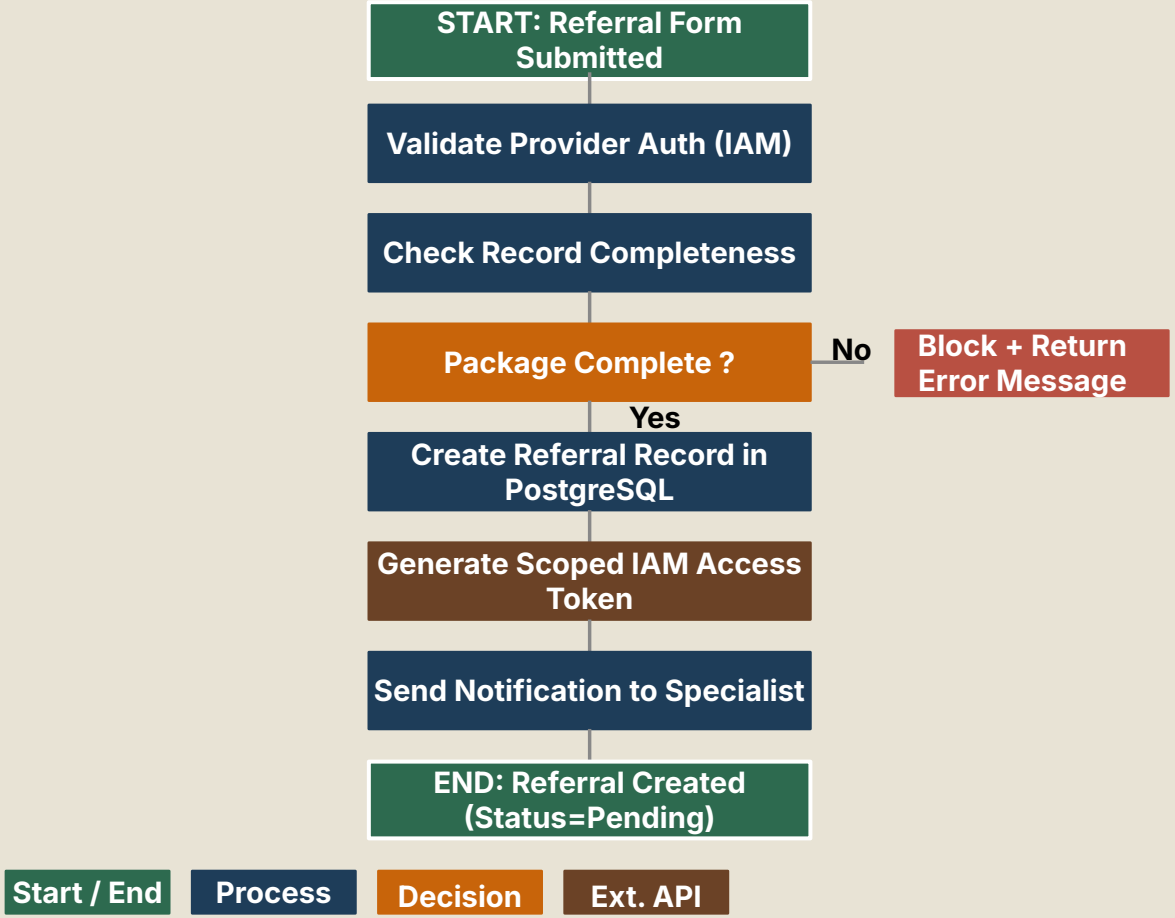
- Provider form submission
- Patient ID, specialist org, record IDs

Outputs

- Referral record in PostgreSQL
- Scoped IAM access token
- Notification to specialist

DB Tables

- Referrals, Referral_Records (junction), Audit_Logs



Algorithm 3: TrackReferralStatus

TrackReferralStatus

User Stories: US-3, US-8

Inputs

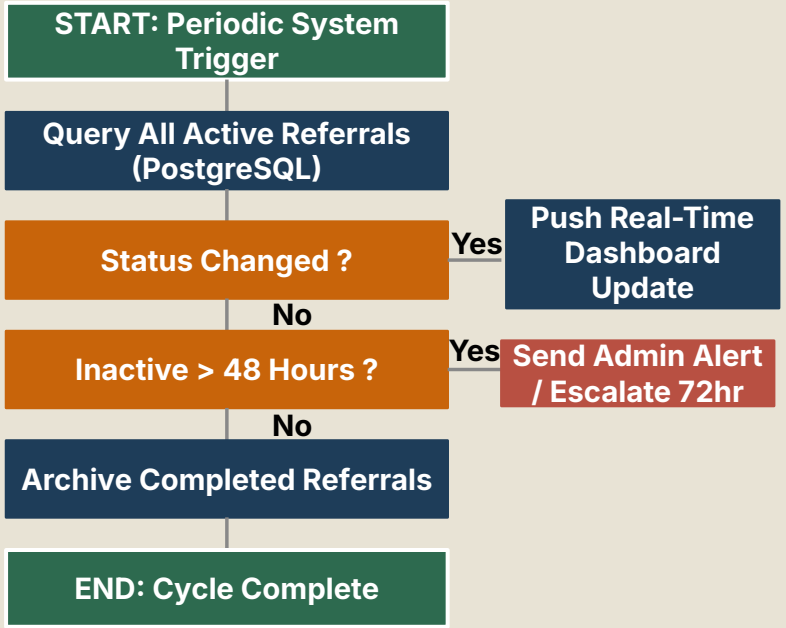
- Periodic system trigger
- Active referral records

Outputs

- Real-time dashboard updates
- Admin alerts (48hr)
- Supervisor escalations (72hr)

DB Tables

- Referrals, Referral_Status_History, Audit_Logs



Start / End | Process | Decision | Ext. API

Algorithm 4: ManageConsent

ManageConsent

User Stories: US-9

Inputs

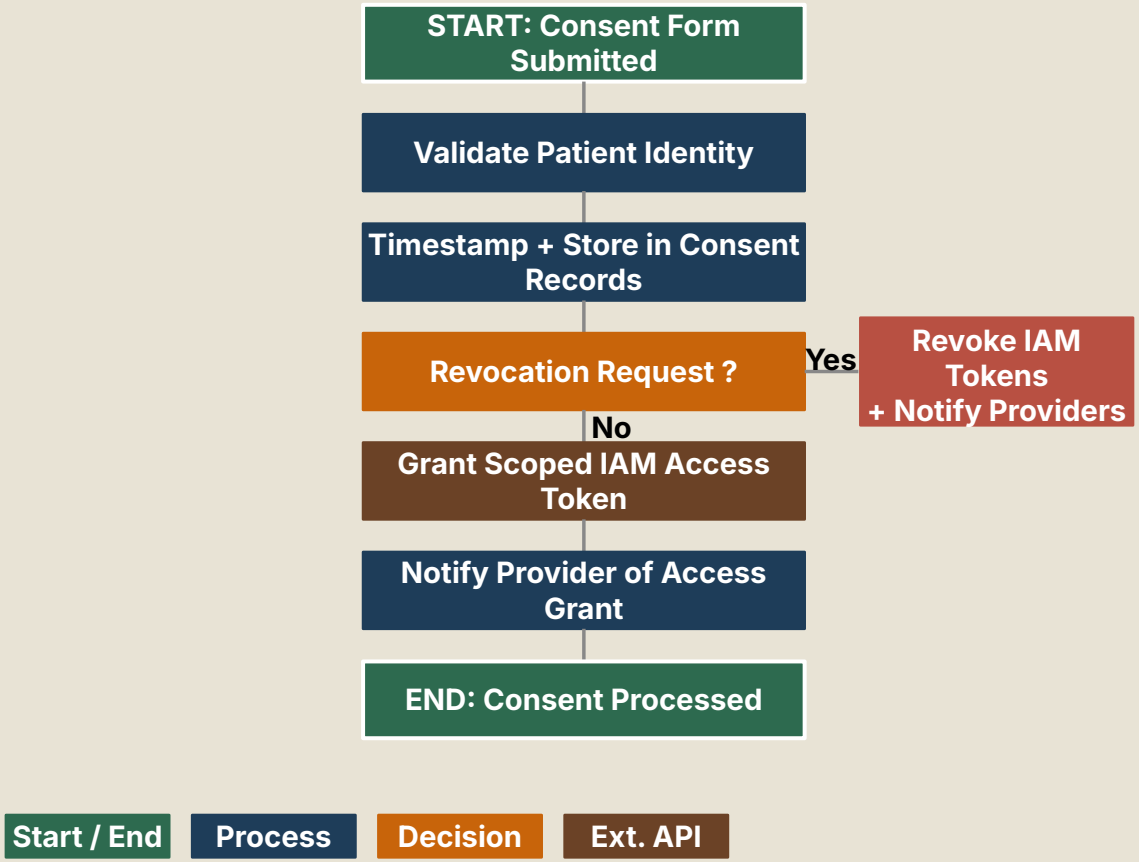
- Patient consent form submission
- Patient ID + provider org

Outputs

- Timestamped consent record
- IAM token grant / revocation
- Provider notification

DB Tables

- Consent, Audit_Logs (revocation events)



Algorithm 5: LogAccessEvent

LogAccessEvent

User Stories: US-7, US-11

Inputs

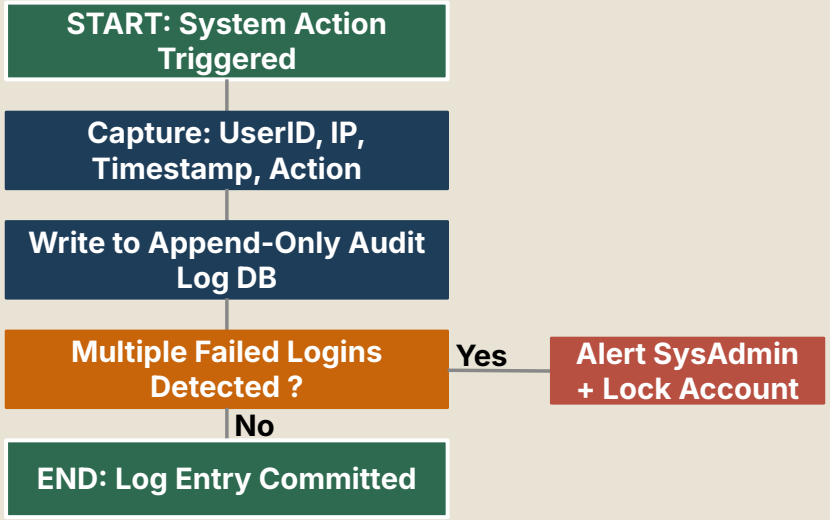
- Any system action (access, transfer, login)

Outputs

- Immutable audit log entry
- SysAdmin alert (if suspicious)

DB Tables

- Audit_Logs (append-only), Users (lockout flag)



Start / End | Process | Decision | Ext. API

Algorithm 6: CacheRecords

CacheRecords

User Stories: US-5

Inputs

- Provider record access request

Outputs

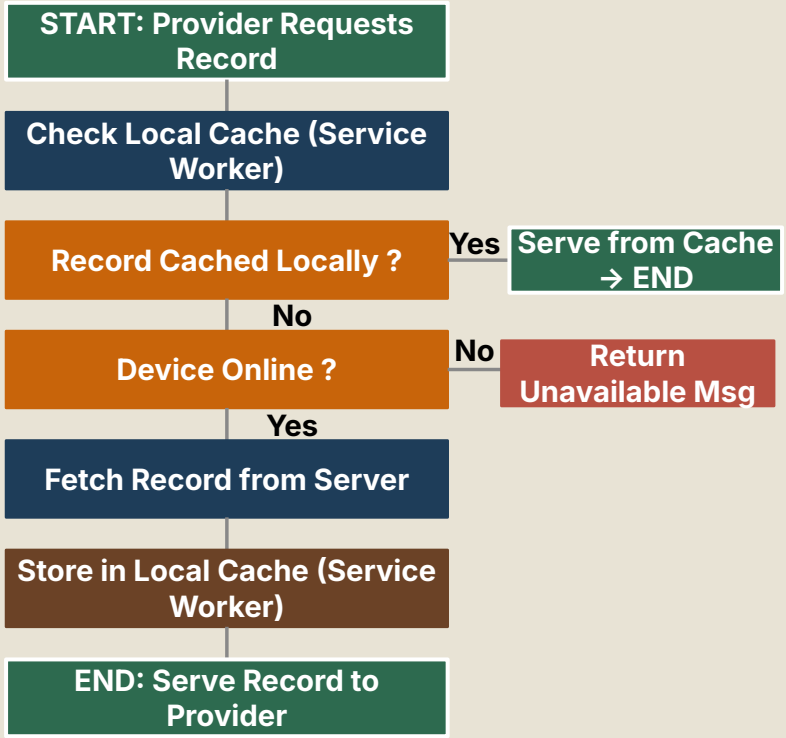
- Record served to provider
- Local cache updated (if online)
- Unavailable message (if offline)

External APIs

- Service Worker (browser cache API)

DB Tables

- Records (read-only, via cache)



Start / End | Process | Decision | Ext. API

Algorithm 7: SyncQueue

SyncQueue

User Stories: US-5

Inputs

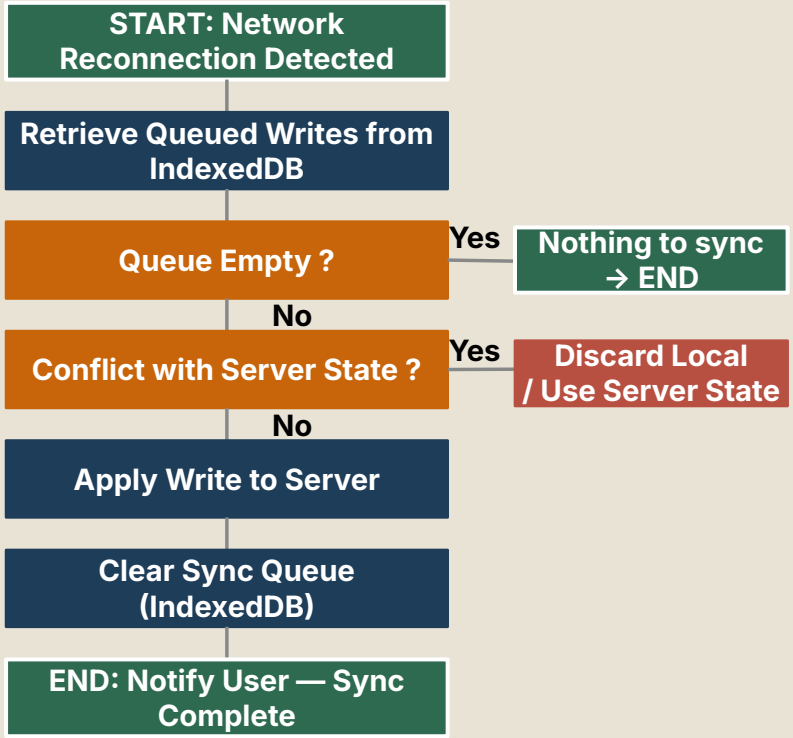
- Network reconnection event
- Local write queue (IndexedDB)

Outputs

- Writes applied to server
- Conflicts resolved (server wins)
- Sync confirmation to user

DB Tables

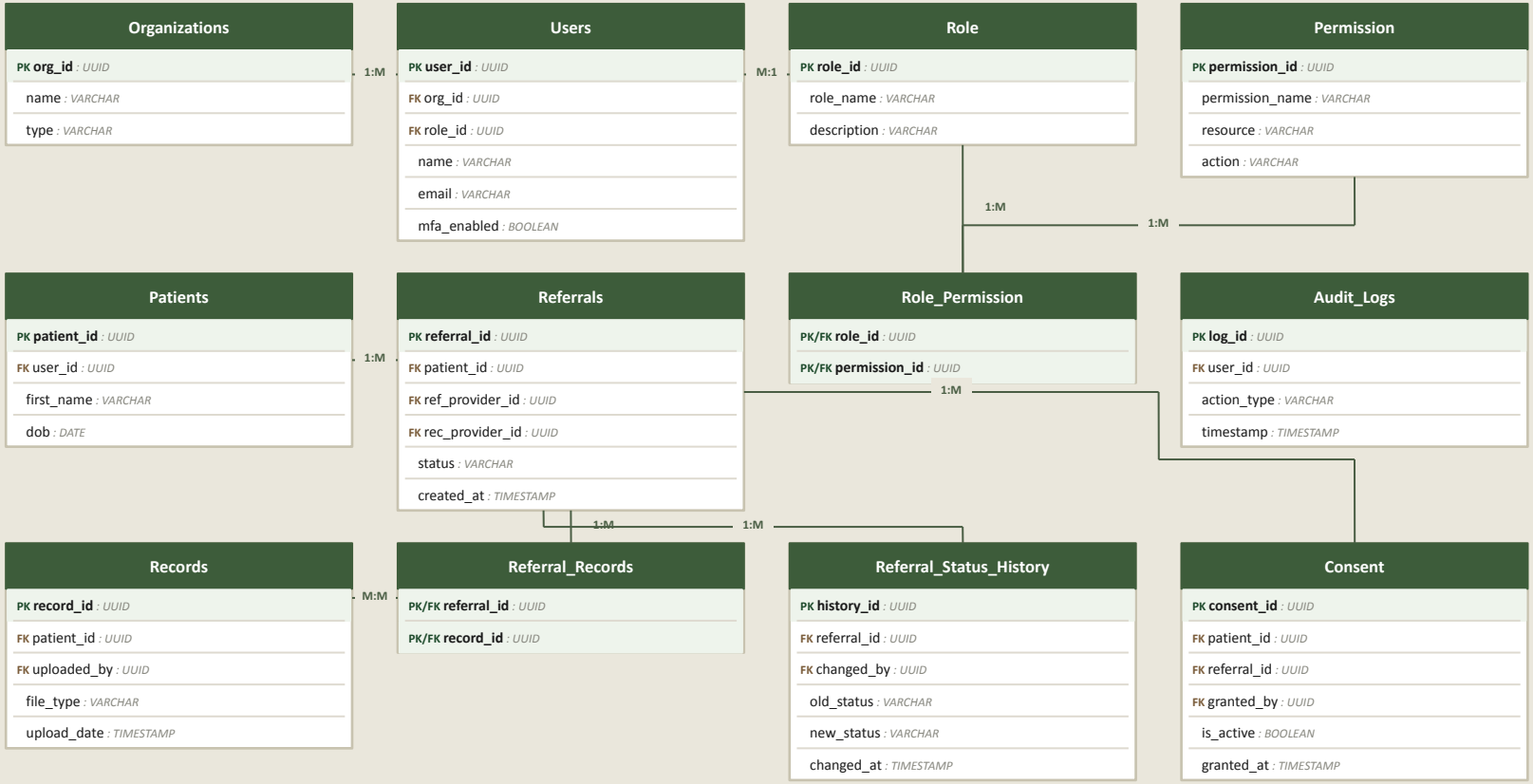
- Records, Referrals (queued writes synced on reconnect)



Start / End | Process | Decision | Ext. API

Database Schema — Entity Relationship Diagram

PK = Primary Key **FK** = Foreign Key field : *TYPE* — = Relationship (1:M · M:M via junction table)



Database Schema — CRUD Mapping to User Stories

Operation	DB Table(s)	Fields / Action	User Story Reference
CREATE	Referrals	referral_id, patient_id, status = Pending	Provider initiates referral in a single workflow
CREATE	Records	record_id, patient_id, uploaded_by, file_type	Provider uploads AES-256 encrypted medical record
CREATE	Consent	consent_id, patient_id, referral_id, is_active = true	Patient provides digital consent for record sharing
CREATE	Role_Permission	role_id, permission_id	Admin assigns role-based access permissions (RBAC)
READ	Records, Consent	All fields via IAM scoped access token	Provider accesses patient records in real-time
READ	Audit_Logs	action_type, user_id, timestamp	Admin exports HIPAA-compliant audit trail (CSV / PDF)
UPDATE	Referrals + Referral_Status_History	Referrals.status updated; history row auto-inserted	Admin monitors and updates active referral status
REVOKE	Consent	is_active = false, revoked_at = CURRENT_TIMESTAMP	Patient revokes record access at any time

Note: REVOKE = soft-delete (UPDATE). Consent rows are never hard-deleted to preserve audit integrity. Referral_Status_History is auto-inserted on every Referrals.status change.

External Interfaces

EHR Systems

- **Standard**: HL7 FHIR R4 API
- **Purpose**: Integration with hospital databases and patient portals.
- **Data In**:
  - Patient Demographics
  - Clinical Observations (Vitals)
- **Data Out**:
  - Updated Referral Status
  - Treatment Documentation

Key Management (Security)

- **Provider:** AWS KMS / Azure Key Vault
- **Purpose:** External root-of-trust for AES-256 encryption.
- **Data In:**
  - Requests for Data Encryption Keys (DEKs) or encrypted metadata
- **Data Out:**
  - Wrapped/ Unwrapped keys for local cryptographic operations.

Identity providers

- **Standard:** OAuth 2.0 / OpenID Connect
- **Purpose:** External authentication via Hospital SSO or Enterprise IdP.
- **Data In:**
  - User Login Credentials (Redirect)
- **Data Out:**
  - Secure JSON Web Tokens (JWT)
  - Validated User Claims/Roles

Development Tools

Development Type	Front-End	Back-End
Language	TypeScript, HTML5, CSS	TypeScript
Framework	React	Node.js + Express.js
IDE	VS Code	VS Code
Version Control	GitHub	GitHub
CI/CD	GitHub Actions	GitHub Actions
Testing Framework	Jest	Jest
API	REST API	REST API
Database		PostgreSQL

Development Tools

Development Type	Front-End	Back-End
ORM		Prisma
Interoperability Standards		FHIR R4, HL7 v2.x
MFA		Speakeasy
Security	Role-Based Access	RBAC, TLS1.3, AES-256
Cloud / Hosting		AWS, Azure, GCP
Key Management		AWS KMS, Azure Key Vault
Offline Support	Service Worker, IndexedDB	Sync Queue
Real-Time Updates		WebSocket

Development Tools

HTML5 & CSS - Standard languages for web page structure and styling

TypeScript - Adds type checking to JavaScript which helps reduce mistakes in healthcare related workflows

VS Code - Easy to use, lightweight IDE that supports both front-end and back-end development

Node.js + Express.js - Used together to build the backend server and handles API requests, logic, and database communications

REST API - API architectural style that allows different software systems to communicate over the internet, it supports core actions like retrieving records, submitting referrals, updating statuses, and managing users

Development Tools

PostgreSQL - Relational database system that stores structured data like users, referrals, and records

Jest - Testing framework that automates unit, integration, and snapshot testing for TypeScript

GitHub - Most commonly used version control platform used to store code, track issues, and allows easy team collaboration through commits, branches, and pull requests

GitHub Actions - GitHub automation tool that runs project tasks whenever code is pushed to make sure the project is stable

React - Framework for building UI with reusable components

Development Tools

Prisma - ORM used to connect the backend to PostgreSQL and simplify database queries

FHIR R4 / HL7 v2.x - Healthcare interoperability standards used to exchange medical data between systems

Speakeasy - Library used to support multi-factor authentication with one-time passcodes

RBAC -Security model that limits system access based on each user’s role

TLS 1.3 - Security protocol used to protect data while it is being sent over the internet

AES-256 - Encryption standard used to protect sensitive data and stored files

Development Tools

AWS / Azure / GCP - Cloud platforms that can host the system and support scalable deployment

AWS KMS / Azure Key Vault - Key management tools used to securely store and manage encryption keys

Service Worker - Browser technology used to support caching and offline access

IndexedDB - Browser database used to store local data for offline support

Sync Queue - Backend process used to save offline changes and sync them when connection returns

WebSocket - Communication technology used to provide real-time updates

Sprint Breakdown

Sprint 0: Project setup

- Initialize GitHub repository and CI/CD pipeline.
- Set up project structure for **TypeScript** and **React**.
- Draft initial **PostgreSQL** schema for users, referrals, and records.
- Configure HIPAA security headers for web server.

Sprint 1: Identity & Access

- **UI:** User can register, log in, and securely access the system.
- **Logic: Node.js + Express.js** service for Role-Based Access Control (RBAC).
- **DB:** Create **PostgreSQL** tables for users and organizational data.

Sprint 2: Secure Record Exchange

- **UI:** Users can upload, view, and retrieve medical records.
- **Logic: Node.js** real-time transfer service using **REST API** architecture.
- **DB:** Securely store medical records and metadata in **PostgreSQL**.

Sprint 3: Interoperability

- **UI: REST API** connection status and data mapping dashboard.
- **Logic: Node.js** FHIR R4 / HL7 Integration Service for EHR compatibility
- **Algorithm: TypeScript** FHIRTranslator module for data normalization.

Sprint Breakdown

Sprint 4: Automated Referrals

- **UI:** Providers can create and send referrals. Users can track referral status in real time.
- **Logic: TypeScript** ReferralValidator module for automated data checks
- **Workflow:** Real-time status tracking via **Node.js** and **REST API**.

Sprint 5: Offline Resilience

- **UI:** Users can continue working without internet.
- **Logic: TypeScript** SyncService for background synchronization.
- **DB:** Utilize browser storage for local caching during connectivity drops.

Sprint 6: Patient Portal

- **UI:** patients can view thier records, and manage consent and data sharing.
- **Logic: REST API** module for secure record release requests.
- **DB: PostgreSQL** tables for consent records and access keys.

Sprint 7: Polish & Finalize

- **Logic:** Finalize immutable audit trail logging in **PostgreSQL**.
- **Testing:** Automated unit and integration testing using the **Jest** framework.
- **UI:** Final **React** dashboard refinement and navigation polishing.

Conclusion

PatientPath modernizes healthcare communication by replacing outdated fax-based processes with a secure, efficient digital platform. By improving access to medical records, reducing administrative burden, and strengthening data security, PatientPath helps providers deliver
faster, safer, and more coordinated care. In a healthcare system where time and accuracy matter, PatientPath provides a necessary step forward.

Glossary

PCP - Primary Care Provider
EHR - Electronic Health Records
HIPAA - Health Insurance Portability and Accountability Act
FHIR - Fast Healthcare Interoperability Resources
RBAC - Role-Based Access Control
HL7 - Health Level Seven

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